

TITLE AND DATE

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HIGH-LEVEL DESCRIPTION:

This is the project document for project 3 of CMPE 311 class spring 2026. It contains all relevant testing requirements for project 3, which extends project 1 and 2 by adding a DC motor-driven fan controlled by a PWM signal. This document outlines the updated customer requirements, high-level technical specifications, system design, and testing procedures and scenarios. It also presents the results of testing.

DESCRIPTION:

This project extends the previous Arduino Uno R3 LED blink-rate controller by adding a DC motor-driven fan. The fan is controlled via a PWM signal generated by the Arduino, with its speed cycled through four states (OFF, LOW, MEDIUM, HIGH, and back) using a single momentary pushbutton. The fan and button operate concurrently with the existing LED serial control system using a round-robin cyclic executive scheduler. This document outlines the customer requirements, corresponding high-level technical specifications, system design, and defined testing procedures.

REFERENCES:

- ProjectAddMotor – The definition of the project this document addresses
- ProjectAsyncTasking_draft – The definition of the previous project
- CMPE310 Project #5 – A similar project assigned in CMPE310 in Spring 2025
- Arduino UNO R3 Product Reference Manual SKU A000066, 12/03/2024
- Async Programming in Arduino: Unleashing the Power of Non-Blocking Code, Mahdi Valizadeh, medium.com, 4/8/2024

DEFINITIONS AND ABBREVIATIONS:

"The User" – The person operating (not programming) the embedded system

"The System" – The embedded system being operated by The User

"The Customer" – The person(s) paying for the embedded system being designed and built

"The Developer" – The person(s) designing and building the System

"The Evaluator" – The person(s) that determine whether or not The System satisfies The Customer-requirements.

"The Customer-requirements" – The requirements defined by The Customer as satisfying The Contract.

"The Requirements" – The System's high-level technical requirements derived from The Customer-requirements.

"The Educational-constraints" – Requirements imposed by the instructor unrelated to the embedded system.

"The Company" – The organization The Customer has contracted with to build The System.

"The Contract" – The business document that legally binds The Company to provide some service or product.

"serial-monitor" – The serial port used by the Arduino IDE to communicate with The User.

"The Reference-platform" – The Arduino compatible ELEGOO Uno R3 development board.

"PWM" – Pulse Width Modulation, a technique used to control motor speed via digital signal.

"MOSFET" – Metal-Oxide-Semiconductor Field-Effect Transistor, used to switch motor power via PWM signal.

"ECR/ECN" – Engineering Change Request/Note, the document describing changes made to the design.

CONVENTIONS:

Must, shall or will – your design must satisfy the requirement

May – your design may satisfy the requirement but doesn't have to

Informative – the intent of the following description is to make the requirement more understandable

All customer requirements are started with "C.#".

All high-level requirements are started with "HL.#".

All testing/validation requirements are started with "T.#"

CUSTOMER REQUIREMENTS:

C1. The User must be able to set the blink rate of two different LEDs.

C2. The User must be able to update the blink rate of each of the LEDs independently.

C3. The LED must blink at the set rate until The User tells the LED to blink at a different rate.

C4. The System must run upon an Arduino Uno R3 compatible development board.

C5. The blink rate of an LED must be expressed in terms of milliseconds.

C6. The device must have a motor-driven fan. (added – Demory, 4/14/2026)

C7. The RPM of the fan must be controllable via a button. (added – Demory, 4/14/2026)

C8. The fan must have 4 speeds: OFF, LOW, MEDIUM and HIGH. (added – Demory, 4/14/2026)

C9. Each hit of the button must cycle the speeds from OFF to LOW, MEDIUM, HIGH, MEDIUM, LOW, OFF. (added – Demory, 4/14/2026)

C10. LOW must be able to move air with minimal sound and minimum feeling of flow. (added – Demory, 4/14/2026)

C11. MEDIUM must be able to move air with moderate sound and a moderate feeling of air flow. (added – Demory, 4/14/2026)

C12. HIGH must be able to move air quickly with moderate sound and a distinct feeling of air flow. (added – Demory, 4/14/2026)

HIGH-LEVEL TECHNICAL REQUIREMENTS:

HL.1. The User through the IDE serial monitor must be able to set the blink rate of two different LEDs. (From requirement C1)

HL.1.1. The blink rate of each LED must be able to be set independent of the other LED. (From requirement C2)

HL.1.2. The setting of an LED blink rate must not interfere with the blinking of the LEDs until the new LED and blink rate are specified. (From requirements C2 and C3)

HL.2. The user through the IDE serial monitor must set the blink rate in terms of once every N milliseconds. (From requirement C5)

HL.3. The System must run upon an Arduino Uno R3 compatible development board. (From requirement C4)

HL.7. The device must have a DC motor-driven fan with an operating voltage between 3V and 12V. (added – Demory, 4/14/2026)

HL.8. The fan must operate at a minimum between 1500 and 6500 RPM. (added – Demory, 4/14/2026)

HL.9. The fan must draw less than 4 Amps. (added – Demory, 4/14/2026)

HL.10. The fan must be driven by a PWM signal with a maximum voltage of 5VDC. (added – Demory, 4/14/2026)

HL.11. The fan must have a maximum dimension less than 4"x4"x3" (not including cage). (added – Demory, 4/14/2026)

HL.12. The RPM of the motor must be controllable using a single-pole, momentary-on, normally-off push button. (added – Demory, 4/14/2026)

HL.13. The operation of the button and motor must have no noticeable effect upon the operation of previously implemented I/O. (added – Demory, 4/14/2026)

HL.14. The fan must have 4 speeds: OFF, LOW, MEDIUM and HIGH. (added – Demory, 4/14/2026)

HL.15. Each hit of the button must cycle the speeds from OFF to LOW to MEDIUM to HIGH to MEDIUM to LOW to OFF. (added – Demory, 4/14/2026)

HL.16. LOW must be able to move air with a sound volume less than 35dB and airflow levels between 30-45 CFM. (added – Demory, 4/14/2026)

HL.17. MEDIUM must be able to move air with a sound volume less than 40dB and airflow levels between 45-70 CFM. (added – Demory, 4/14/2026)

HL.18. HIGH must be able to move air with a sound volume less than 50dB and airflow levels between 70-100 CFM. (added – Demory, 4/14/2026)

TESTING AND VALIDATION REQUIREMENTS:

- T.1 – Dialogue from serial monitor sample matches Table 4.
- T.2 – Blink rate of each LED is independent and doesn't change until prompted.
- T.3 – Blink rate is specified in milliseconds.
- T.4 – Blink rate must be correct.
- T.5 – The blink rate setting must be able to repeat at least 5 times.

- T.6 – Button press sequence matches Table 5: OFF > LOW > MEDIUM > HIGH > MEDIUM > LOW > OFF. (added – Demory, 4/14/2026)
- T.7 – Fan speed changes are reflected correctly in the serial monitor after each button press. (added – Demory, 4/14/2026)
- T.8 – Fan operation does not interfere with LED blinking or serial monitor dialogue. (added – Demory, 4/14/2026)
- T.9 – Fan LOW speed produces minimal airflow and sound. (added – Demory, 4/14/2026)
- T.10 – Fan MEDIUM speed produces moderate airflow and sound. (added – Demory, 4/14/2026)
- T.11 – Fan HIGH speed produces distinct airflow. (added – Demory, 4/14/2026)
- T.12 – Button sequence must be able to repeat at least 3 full cycles. (added – Demory, 4/14/2026)

DESIGN:

DESIGN PRE-REQUISITES:

- ELEGOO Arduino Uno R3 clone
- Arduino IDE 2.3.3 or better
- Breadboard
- Two LEDs
- DC motor / fan
- N-channel MOSFET (e.g. IRF540N)
- 1N4007 flyback diode
- 220Ω resistor (gate), 10kΩ resistor (pull-down)
- 100μF capacitor (smoothing)
- 9V battery + connector
- Momentary pushbutton
- Connective wires

DEVELOPMENT PLATFORM:

- Uses Arduino IDE
- Codes in C++

Code is attached in submission and located on GitHub repository.

Video is attached to submission.

TESTING:

Test Performed	Results
T1	Satisfied – see video
T2	Satisfied – see video
T3	Satisfied – see video
T4	Satisfied – see video
T5	Satisfied – see video
T6	Satisfied – see video
T7	Satisfied – see video
T8	Satisfied – see video
T9	Satisfied – see video
T10	Satisfied – see video
T11	Satisfied – see video
T12	Satisfied – see video

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